Deep Etching of $LiNbO_3$ Using Inductively Coupled Plasma in SF_6-Based Gas Mixture

Artem A. Osipov, Armenak A. Osipov, Gleb A. Iankevich, Anastasiya B. Speshilova, Alexander Shakhmin, Vladimir I. Berezenko, and Sergey E. Alexandrov

***Abstract*— This work is an extensive study of the plasma chemical etching (PCE) process of single-crystalline lithium niobate ($LiNbO_3$) in the SF_6/O_2 based inductively coupled plasma (ICP). The influence of the main technological parameters of the $LiNbO_3$ PCE process, including the distance between the sample and the lower edge of the discharge chamber, as well as the temperature of the substrate holder on the etching process rate, has been studied. It was shown that changing the temperature of the substrate holder in the range from 100 to 200 °C leads to a gradual rise of the etching rate from 127 to 282 nm/min. A further increase in the temperature to 250 °C results in a sharp increase in the etching rate to 711 nm/min. The maximum achieved etching rate in experimental series which were aimed at determining the dependence of the $LiNbO_3$ etching rate on the temperature of the substrate holder was 812 nm/min at a substrate holder temperature of 325 °C. With the help of X-ray photoelectron spectroscopy (XPS) technique was found and shown that during the etching process in fluorinated plasma the nonvolatile LiF compound is formed on the surface of the treated $LiNbO_3$. On the basis of the obtained results, the optimal process of deep (>80 μm) $LiNbO_3$ PCE was developed, with an etched wall inclination angle of ≈ 110°, with selectivity ratio to Cr mask of ≈ 20, and an etching rate of about 300 nm/min. [2020-0317]**

***Index Terms*— Heating, inductively coupled plasma, plasma etching, lithium niobate, SF_6/O_2 plasma.**

I. Introduction

MONOCRYSTALLINE lithium niobate ($LiNbO_3$) is the basis material for the production of different devices of the optical industry, including the microwave photonic branch, which is in the focus of research activities at the moment [1]–[3]. In recent years, a whole class of functional and digital photonic integrated circuit (PIC), such as switching matrices, optical spectrum analyzers, microwave phase and amplitude modulators, holographic memory elements, biomedical devices, as well as sensors of physical quantities have been implemented based on $LiNbO_3$ crystals [4]–[8]. The reason why $LiNbO_3$ became a core material of modern nonlinear optics, electro optics and acoustoelectronics is because of its high values of electro-, acousto- and nonlinear optical coefficients [9]–[13]. Besides, the devices made on the basis of $LiNbO_3$ have found wide application in automobile and oil industries, and also in aircraft engine production [14]. The crucial technological operation at production of such devices is the creation of structures of different patterns (holes, grooves, etc.) in/on the $LiNbO_3$ substrates. Such problems can be solved by plasma “dry” and liquid etching techniques. The main disadvantages of liquid etching are: low etching rates, isotropic nature of etching and low automation of the etching process [15]–[17]. The above-mentioned disadvantages do not allow the formation of deep vertical structures in lithium niobate. An interesting technique of deep structures formation in $LiNbO_3$ was proposed by the authors [18]. For creation of deep vertical structures, they used a diamond disk which leaded to high values of the anisotropy. However, it has a number of drawbacks such as: low productivity, limitation on the creation of different shapes and sizes structures due to the original shape of the diamond disk, as well as the introduction of contaminants (abrasive) on the samples surfaces. Plasma etching has a number of benefits such as: possibility of the process automation, and the ability to form highly oriented structures in $LiNbO_3$. Many works have been devoted to $LiNbO_3$ etching processes in SF_6, CHF_3, C_4F_8, CF_4, BCl_3 based plasma and their mixtures with $Ar/N_2/O_2/H_2$ [6], [19]–[22]. In the vast majority of studies aimed at the different structures creation on the surface of $LiNbO_3$ substrates at the power of the RF inductively coupled plasma source up to 1000 W the etching rates are in the range from 50 to 150 nm/min [8], [12], [13], [23]. Further increase of etching rate up to 350 nm/min was achieved due to increase of RF power supplied to plasma discharge up to 2000 W [24], [25]. Typical depths of etching areas achieved by ICP etching of $LiNbO_3$ are lying in the range from 500 nm to 10 μm [11], [24]–[31]. Thus, the achievement of the high etching rates values is important and is possible only with the right choice of a combination of a specific process parameter, such as the gas mixture composition, which is defined both by the type of gases used and by gases flow rates (Q_x, x - name of the gas) of individual gas component, pressure (P) in the reaction chamber, substrate holder HF

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Artem A. Osipov is with the Russian Academy of Sciences, Academic University, 194021 Saint Petersburg, Russia (e-mail: tema.osipov@mail.ru).

Armenak A. Osipov is with the Southern-Urals Federal Research Center of Mineralogy and Geoecology, Institute of Mineralogy, Ural Branch of the Russian Academy of Sciences, 456317 Miass, Russia.

Gleb A. Iankevich is with the Institute of Nanotechnology, Karlsruhe Institute of Technology, 76344 Eggenstein-Leopoldshafen, Germany.

Anastasiya B. Speshilova, Alexander Shakhmin, Vladimir I. Berezenko, and Sergey E. Alexandrov are with the Higher School of Physics and Materials Engineering, Peter the Great St. Petersburg Polytechnic University, 195251 Saint Petersburg, Russia.

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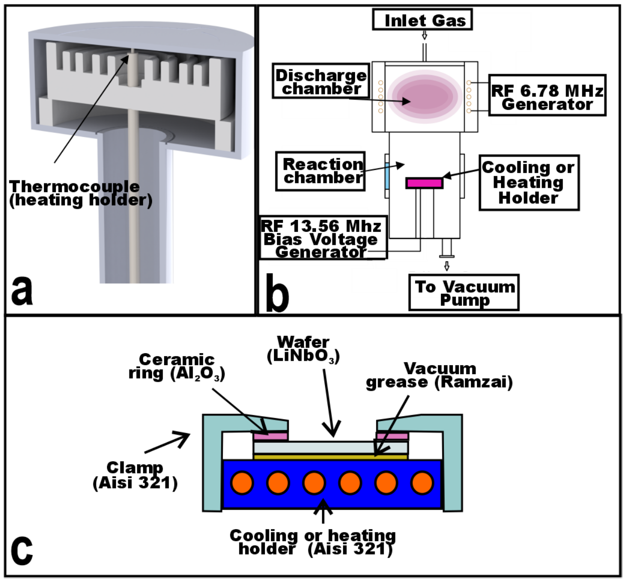

Fig. 1. (a) 3D CAD drawing of the substrate holder with a built-in heater, (b) schematic drawing of the processing and plasma chambers [32], [33], (c) schematic representation of sample position on substrate holder during the etching process.

electrode bias voltage (U_{bias}), etc. However, the main problem in the formation of deep structures (more than 10 μm) is the generation of a non-volatile LiF compound on the etching surface, which slows down the plasma chemical etching (PCE) process, and with increasing process time can lead to a complete stop of the etching process [24]. Thus, the question of achieving deep structures (more than 10 μm) in $LiNbO_3$ is very relevant, but poorly studied.

II. Experimental Details

The experiments were carried out on a custom-designed plasma chemical etching system with an inductively coupled high density plasma source (Figure 1-b). For experiments at different temperatures, the system was equipped with a substrate holder with an internally integrated resistive heating unit as well as water cooling unit (the temperature of running water is 15°C). The design of the substrate holder made it possible to heat and hold the required sample temperature in the range of between 20 and 400 °C. The body of the substrate holder is made of corrosion-resistant AISI 321 stainless steel. Since the direct control of the substrate temperature under plasma etching conditions is itself a technically challenging task, the substrate holder surface temperature was measured in this work. The control was carried out using a type-K thermocouple (Figure 1-a) located in close proximity to the lower plane of the holder top plate, to which a $LiNbO_3$ sample was mounted.

As can be seen from Figure 1-b, the reactor of the system is made of two chambers - discharge and reaction. The plasma in the discharge chamber (inner diameter of the discharge chamber is 23 cm, and h = 22.5 cm) is created by supplying high-frequency (HF) power to the inductor of a specific geometry from the HF generator (f = 6.78 MHz, W_{max} = 1000 W) through a resonant matching device. The cylindrical shape reaction chamber (inner diameter of the reaction chamber is 23 cm, and h = 33.5 cm) is made of stainless steel AISI 321. To form the bias potential, a 13.56 MHz HF voltage was applied to the substrate holder (electrode) from a separate HF generator. As samples for etching XY-128° cut $LiNbO_3$ were used with thickness of 370 microns. Metal film made of Ni-Cu-Cr thickness of 7 microns with linear dimensions of windows 3×10 mm was used as a mask. The sample size was 24×24 mm. In all experiments the samples were cleaned with acetone, ethyl alcohol and deionized water for 10 minutes consecutively in an ultrasonic bath before the PCE process. In addition, after loading a sample into the reaction chamber and before the PCE experiment, the samples were treated in argon plasma for 10 minutes to remove residual contaminants from the substrate surface. The parameters of the cleaning process in argon were as follows: applied power (W = 750 W), pressure in the reaction chamber (P = 0.75 Pa), Ar gas flow (Q_{Ar} = 21.75 sccm), and the distance between the bottom part of the discharge chamber and the surface of the sample (h = 15 cm). An important property of lithium niobate is the generation of elastic stresses in the crystal due to the occurring temperature gradient, which can lead to the sample cracking [34]–[36]. In this regard, to improve heat transfer from the holder to the sample, the $LiNbO_3$ sample was mounted with the vacuum lubricant "Ramzai" (TU 38.5901248-90). A ceramic O-ring with an outer diameter of 50 mm and an inner diameter of 15 mm was placed on top of the sample. On top of the ceramic ring a steel cap (AISI 321) with a 15 mm diameter opening was placed to press the sample to the holder. Thus, the active process area (etching area) was a 15 mm diameter circle. Due to the fact that the basis of the plasma chemical etching process is the reaction of interaction of fluorine radicals with $LiNbO_3$ surface, the possible etching reaction can be presented in the following form:

$$LiNbO_3 + F^* \rightarrow LiF \downarrow + NbF_x \uparrow + O_2 \uparrow + OF_2 \uparrow + \ldots \quad (1)$$

Thus, high purity sulphur hexafluoride SF_6 99.998% (GOST TU 6-02-1249-83) was used as the main etchant gas. In terms of increasing the etching rate, oxygen (O_2) (high purity, TU 2114-001-05798345-2007) was used as an additive gas for choosing an optimal gas mixture. The etching process was initiated when the surface temperature of the substrate holder reached the desired value. After process the etching depth was measured. To ensure the repeatability of the etching process, all experiments were performed three times each. Etching depth was estimated with the help of microphotographs obtained on the CarlZeiss Supra 55VP scanning electron microscope with an accuracy of ±2.5 %.The etching depth was determined as the difference between the averaged values of the sample thickness before and after process. The etching rate (V_{etch}) was calculated as the ratio of the etching depth to the time during which the etching process was performed. The elemental composition of the surface of the $LiNbO_3$ after the etching process was investigated by X-ray photoelectron spectroscopy on a cluster type system "Nanofab-25".

TABLE I
PARAMETERS OF THE CONTROL EXPERIMENT

W, W	Q_{SF6}, sccm	Q_{O2}, sccm%	P, Pa	U_{bias}, V	T, °C	h, cm
550	10.15	3.45	1.65	-50	15	5

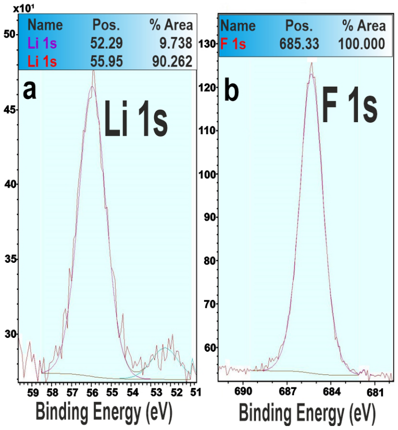

Fig. 2. XPS of etched window surface registered after PCE process (a - Li, b - F).

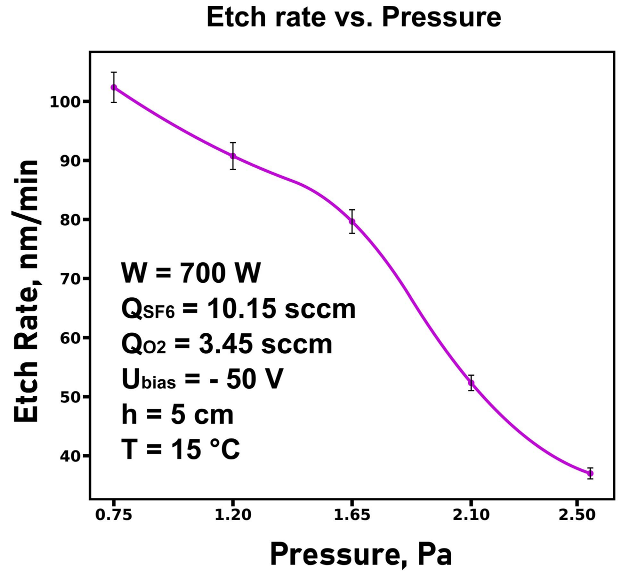

Fig. 3. Plot of dependence of $LiNbO_3$ etching rate on pressure in reaction chamber.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

In the preliminary series of experiments the ratio $[Q_{O2}/(Q_{O2}+Q_{SF6})\cdot 100\%$ was ≈25%. The parameters of the experiment are presented in Table I. The etching time was 30 minutes.

After the etching process, black deposits were observed on the entire surface of the processed $LiNbO_3$ area. The results of the elemental composition investigation of the $LiNbO_3$ etching surface showed that during the process in SF_6/O_2 plasma a non-volatile LiF compound is formed on the treated surface. In the lithium spectrum there are 2 components (Figure 2-a, the largest of which corresponds to LiF (55.95 eV), and the second, weak (52.29 eV), corresponds to $LiNbO_3$. The fluorine line contains one component, which corresponds to LiF (685.3 eV) (Figure 2-b). The average etching rate of $LiNbO_3$ in this series of experiments was 61.2 nm/min. It was found that an effective way to remove LiF from the etched surface was to treat $LiNbO_3$ samples after the PCE process in hydrofluoric acid (40%) for 30 seconds.

With the aim of investigation of physical and chemical bases of plasma etching, the influence nature of the etching process technological parameters on etching rate of $LiNbO_3$ is investigated. In all experiments after the PCE process the samples were treated in 40% hydrofluoric acid for 30 seconds to remove non-volatile LiF. During the experimental series, the W_{RF} varied from 550 to 700 W, P from 0.75 to 2.55 Pa, U_{bias} from -50 to -150 V with a step of 25 V, the concentration of O_2 in the initial gas mixture ranged from 15 to 35% ($[Q_{O2}/(Q_{O2}+Q_{SF6})\cdot 100\%$), h varied from 5 to 15 cm and T varied from 100 to 325 °C. The Q_{SF6} was fixed at 10.15 sccm. First of all, of particular interest to the research are the parameters (P, U_{bias}, h) that can most strongly affect the ionic sputtering of $LiNbO_3$ substrate. Physical component of the etching process causes uncontrolled and uneven heating of the plate, which can result in mechanical destruction of the $LiNbO_3$ substrate. As shown in many studies [6], [37]–[40], the pressure in the reaction chamber significantly affects the intensity of ionic bombardment of the substrate surface during the plasma etching of various materials. Therefore, first of all, the influence of this parameter on the etching rate was studied. All other technological parameters were fixed. The obtained results are shown in Figure 3 as V_{etch}(P) dependence. Etching time in all experiments was 30 minutes.

Most likely, the increase in the etching rate of $LiNbO_3$ with a decrease in pressure is due to an increase in free path length and energy of ions that bombard the surface of the processed material [40]. Thus, the reactor pressure is a parameter that directly influences the contribution of the physical part of the plasma etching process of $LiNbO_3$ in the selected range of technological parameters. It should be noted that during these experiments no mechanical destruction of $LiNbO_3$ samples was detected in any case in the whole studied pressure range. As can be seen from Figure 4, the etching rate increases almost linearly with increasing bias voltage. The increase in bias voltage leads to an increase in the energy of ions bombarding the sample surface, which causes an increase in the etching rate of $LiNbO_3$ by intensifying the physical component of the PCE process.

It was also found that samples exposed to U_{bias} values, varying from -100 to -150 V, either cracked or became brittle. This indicates the generation of significant temperature gradients in the sample when large bias voltage values are used. The intensity of ion bombardment must also be affected by the distance between the substrate and the lower edge of the discharge chamber, i.e., the distance from the surface of the material being processed to the plasma source.

As you can see from Figure 5, the $LiNbO_3$ etching rate increases as the substrate holder approaches the discharge chamber. It can be assumed that as the sample approaches the plasma generation zone, the total number of ions bombarding the etching surface increases. Another important parameter that significantly affects the plasma chemical etching process is the concentration of oxygen added to the main etching

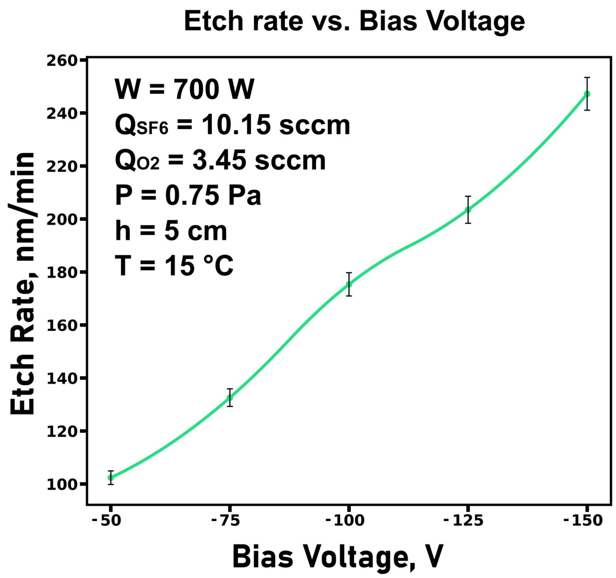

Fig. 4. Dependence plot of $LiNbO_3$ etching rate on the bias voltage applied to the substrate holder.

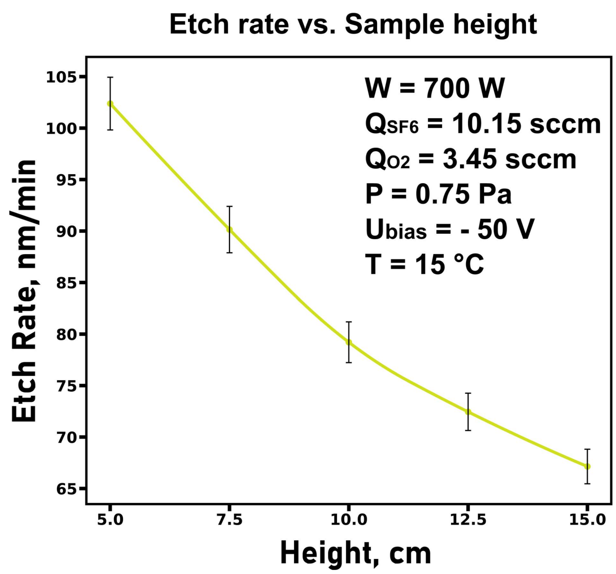

Fig. 5. Plot of dependence of $LiNbO_3$ etching rate on the distance between the substrate holder and the bottom edge of the discharge chamber.

gas [41]–[43]. The dependence curve of $V_{etch}(Q_{O2})$ is shown in Figure 6.

It follows from the obtained data (Figure 6) that increasing the concentration of oxygen from 15% increases the etching rate by 20% and V_{etch} it reaches its maximum value at 25% (102 nm/min). The increase in etching rate is most likely due to an increase in the concentration of F* radicals in the discharge, as well as the possible acceleration of chemical reactions that produce oxygen-containing volatile chemical compounds ($NbOF_3$, OF_2). However, with further increase of O_2 concentration, most likely, there is a significant dilution of the main etching gas occur [32], [44], which leads to a decrease in the concentration of chemically active plasma particles (CAP) and, consequently, to a decrease in the etching rate by more than 40%. Another parameter affecting the $LiNbO_3$ etching rate is the substrate holder temperature. Previously we already investigated and discussed the influence

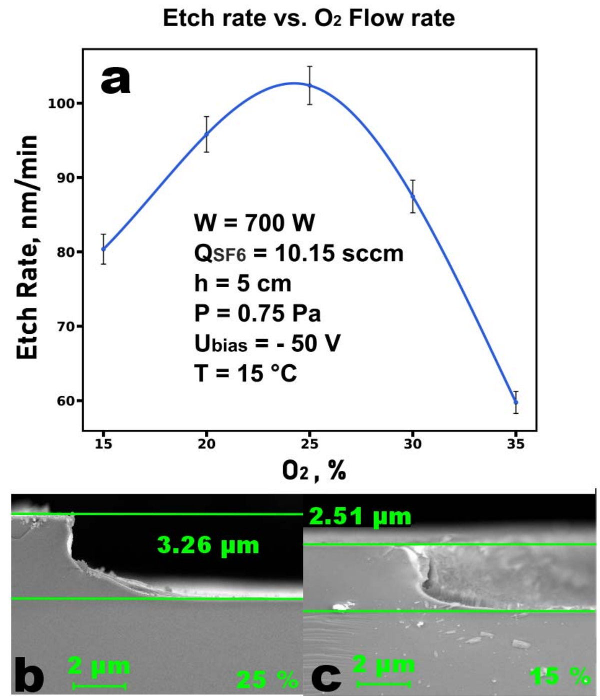

Fig. 6. (a) Dependence of the $LiNbO_3$ etching rate on the oxygen concentration in the total gas mixture ($[Q_{O2}/(Q_{O2} + Q_{SF6}) \cdot 100\%$), (b) microphotograph of the etched profile ($Q_{O2} = 25\%$), (c) microphotograph of the etched profile ($Q_{O2} = 15\%$).

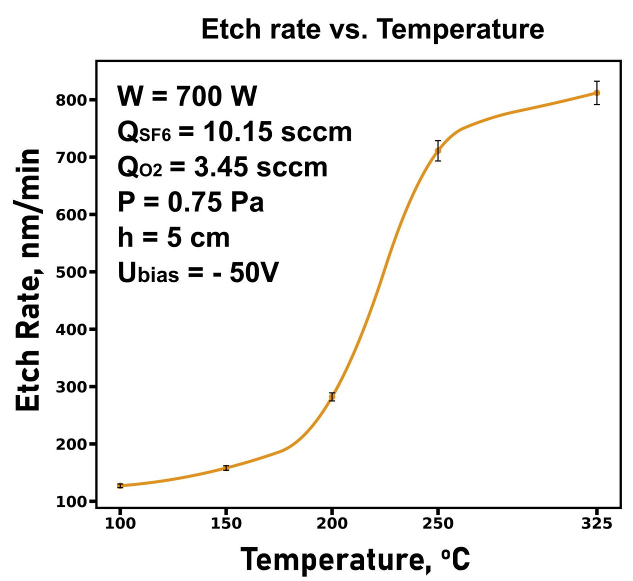

Fig. 7. The dependence plot of $LiNbO_3$ etching rate on the temperature of the substrate holder [33].

of the substrate holder temperature on the etching rate [33]. The result of this experimental series is presented in Figure 7. It can be seen that the obtained $V_{etch}(T)$ dependence has a complex nature, however, in general, the temperature increase gives a significant increase in etching rate. It is likely that such a strong increase in the $LiNbO_3$ etching rate is due to an increase in the chemical reactions rate on the surface of the sample. Probably, additional transfer of energy due to heating of the substrate holder and, consequently, the sample increases the etching rate due to NbF_5 removal, the boiling point of which is 235 °C [45].

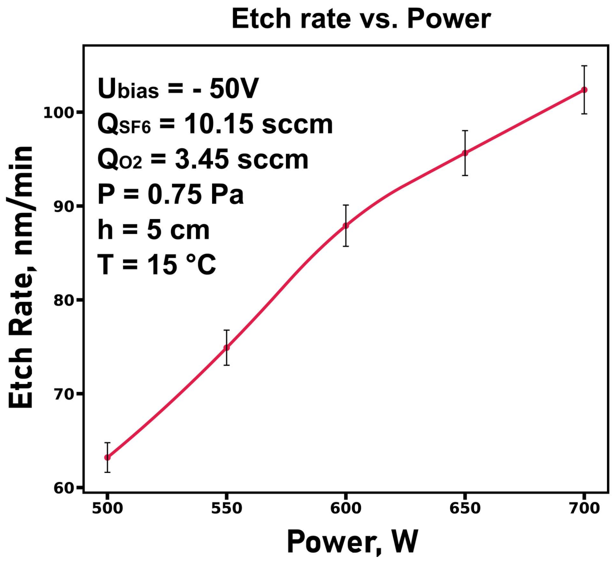

Fig. 8. Dependence plot of $LiNbO_3$ etching rate on RF source power.

A significant role in the PCE process is held by the applied RF power, in this regard an experimental series aimed at determining the dependence nature of the $LiNbO_3$ etching rate on the inductively coupled plasma RF power source was conducted.

As can be seen from the results shown in Figure 8, with increasing RF power, the etching rate of $LiNbO_3$ increases by a nearly linear trend. Such nature of dependence, probably, is connected with the fact that with increasing RF power intensity of inelastic collisions with electrons increases, due to which CAP concentration in the discharge increases [32], [40]. After analyzing the obtained data, we can conclude that plasma-chemical etching of $LiNbO_3$ happens due to both chemical reactions (chemical part of the etching process) and ionic sputtering of the substrate (physical part of the etching process). However, it is important to notice, that at temperature up to $\approx 200\,^{\circ}C$ the greatest contribution is the physical component of process what is evidenced by to what strong dependence of $LiNbO_3$ etching rate on bias voltage. It shows - the rate increases proportionally to the rise of the bias voltage nearly in 2.5 times and goes up to 250 nm/min at 150V (Figure 4), in the given range of technological parameters. Further temperature increase leads to the following - chemical component of the process makes the largest contribution to $LiNbO_3$ PCE, which is confirmed by the dependence of V_{etch} on T (Figure 7). Based on the analysis of the obtained experimental data a series of control experiments on deep, high-rate, directed $LiNbO_3$ substrates PCE was conducted. Process was carried out in several stages, every 30 minutes the sample was cleaned from LiF in hydrofluoric acid, the treatment time was 30 seconds. Then the sample was treated in distilled water for 10 minutes, after which the etching process was resumed. Technological parameters of the control series of experiments are presented in Table II.

In order to increase the etching rate, it was decided to increase the bias voltage to -200 V in absolute values. In order to avoid destruction of the sample from uncontrolled heat caused by ion bombardment, the sample was moved away from the lower edge of the discharge chamber by 15 cm.

After the etching process, the sample did not crack, so it can be concluded that the key parameter that affects the uneven temperature distribution is the distance between the sample and the plasma generation zone for the selected range of process parameters. As can be seen from Figure 9, the wall inclination angle of the processed opening was 110°, probably is due to erosion of the metal mask caused by high process temperature and high-energy ions [46], [47]. Process final etching rate was about 305 nm/min, with etching depth over 80 μm and selectivity to Ni-Cu-Cr mask about 20. The decrease in etching rate in comparison to previous experiments (dependence of etching rate on the temperature of the substrate holder) is probably due to the increased duration of etching process, which leads to the formation of thicker and denser LiF layer, which slows down the PCE process.

TABLE II

TECHNOLOGICAL PARAMETERS OF THE CONTROL EXPERIMENT $LiNbO_3$ PCE PROCESS

W, W	Q_{SF6}, sccm	Q_{O2}, sccm%	P, Pa	U_{bias}, V	T, °C	h, cm	t, min
700	10.15	3.45	0.75	-200	325	15	270

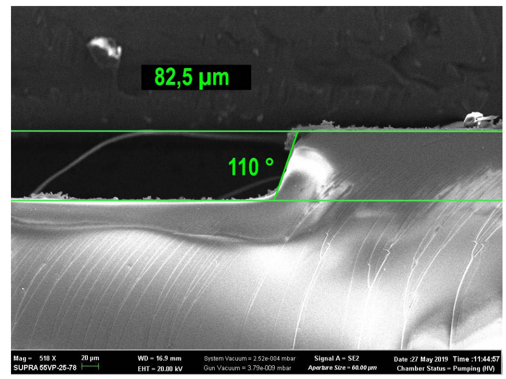

Fig. 9. Microphotograph of the etched profile in the check experiment.

IV. CONCLUSION

As a result of this study with the XPS, it has been demonstrated that the non-volatile LiF compound formed on the surface during the PCE process which slows down the etching process. It was found that an effective method of removing LiF from the surface of processed $LiNbO_3$ was to treat the sample in 40% hydrofluoric acid. The experiments aimed at determining the $LiNbO_3$ etching rate dependence on technological process parameters demonstrated that in the substrate holder temperature range up to $200\,^{\circ}C$ the physical component has a greatest contribution to the etching process (i.e. ion bombardment). At higher temperature values, there is a sharp rise in the $LiNbO_3$ etching rate up to 812 nm/min and the chemical component of the process already makes greater contribution to the etching process. It is likely that the increase in etching rate with the temperature rises above $200\,^{\circ}C$ is due to the removal of NbF_5 from the processed opening surface, as the boiling point of this compound is 235 °C. Thus, as a result of this work, we were able to achieve a record etching depth (82.5 μm) in $LiNbO_3$ with PCE in SF_6/O_2 plasma. Etching rate was about 306 nm/min, at 700 W RF power, with selectivity ratio to Ni-Cu-Cr mask ≈20. The inclination angle of the processed opening wall was about 110°, which may be due to erosion of the metal mask.

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